

ZHANGJIAKOU, China

WMO: 544010

Lat: 40.7694N

Lon: 114.9194E

Elev: 2540

StdP: 13.40

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.0	(c) 4.4	(d) -21.7	(e) 1.8	(f) 11.0	(g) -17.8	(h) 2.3	(i) 12.7	(j) 19.0	(k) 19.8	(l) 16.8	(m) 17.2	(n) 7.0	(o) 340	(p) 0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.3	(c) 91.6	(d) 66.1	(e) 88.8	(f) 65.5	(g) 86.2	(h) 65.3	(i) 73.1	(j) 83.3	(k) 71.7	(l) 81.5	(m) 70.1	(n) 79.8	(o) 5.9	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	121.7	78.7	68.5	115.1	77.7	66.9	108.6	76.4	38.7	83.4	37.3	81.7	35.9	79.4	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
17.5	14.6	12.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-4.0	96.1	4.8	3.6	-7.4	98.7	-10.2	100.7	-12.9	102.7	-16.4	105.3		
			-5.3	75.6	4.4	2.1	-8.5	77.1	-11.1	78.3	-13.6	79.5	-16.8	81.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	18.1	25.3	38.3	53.0	64.5	72.3	76.3	73.5	64.1	50.4	34.3	21.5
	DBStd	21.53	8.66	8.18	8.68	8.26	6.95	5.50	4.47	4.32	6.09	7.07	8.36	6.99
	HDD50	3553	990	691	373	61	2	0	0	0	80	471	885	
	HDD65	6675	1455	1111	828	367	94	8	1	2	86	453	920	1350
	CDD50	3343	0	0	11	151	451	670	815	728	422	94	1	0
	CDD65	987	0	0	0	7	77	228	351	265	58	1	0	0
	CDH74	8474	0	0	1	112	849	2045	3066	1944	445	12	0	0
	CDH80	2841	0	0	0	20	257	746	1152	575	91	0	0	0
Wind	WSAvg	5.4	5.7	5.8	6.0	6.3	6.2	5.2	4.8	4.7	4.8	4.9	5.1	5.5
Precipitation	PrecAvg	15.7	0.1	0.2	0.4	0.6	1.2	2.2	4.2	4.2	1.7	0.7	0.2	0.0
	PrecMax	23.3	0.4	1.0	1.3	2.8	4.5	4.2	12.3	10.7	4.3	2.7	0.8	0.2
	PrecMin	8.9	0.0	0.0	0.0	0.0	0.2	0.9	0.6	1.3	0.4	0.0	0.0	0.0
	PrecStd	3.7	0.1	0.2	0.3	0.6	0.8	0.7	2.3	2.2	0.8	0.6	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	51.0	69.4	82.9	91.3	95.2	95.8	92.6	86.9	75.4	59.7	42.3
		MCWB	29.9	35.9	46.2	55.1	59.9	63.7	68.5	67.9	63.3	56.7	43.7	31.2
	2%	DB	35.8	46.1	62.4	77.0	86.4	90.8	92.2	88.2	82.5	70.2	54.7	38.3
		MCWB	26.8	33.5	43.3	52.3	58.3	63.4	68.4	67.4	62.2	52.4	41.1	29.1
	5%	DB	32.4	42.4	57.7	72.6	82.4	87.4	89.1	85.6	79.0	66.4	50.5	35.1
		MCWB	24.5	31.3	40.8	50.3	56.7	62.6	68.0	67.1	60.6	50.7	38.9	27.0
	10%	DB	29.2	38.8	53.1	68.3	78.6	84.3	86.3	83.2	75.7	62.8	46.8	32.1
		MCWB	22.3	29.3	38.4	48.5	55.3	62.3	68.0	66.7	59.0	49.5	37.1	24.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.4	37.9	48.0	58.4	65.0	70.2	76.1	74.4	68.7	59.1	45.6	32.3
		MCDB	40.0	47.3	65.6	73.1	81.6	84.1	86.7	84.7	80.3	69.7	53.5	40.7
	2%	WB	27.2	34.2	44.4	54.9	62.0	68.5	73.8	72.7	65.7	56.0	42.7	29.5
		MCDB	35.2	44.2	60.4	71.4	77.7	81.4	84.7	81.9	75.1	65.3	52.7	37.8
	5%	WB	24.7	31.8	41.8	52.4	60.1	67.0	72.4	71.2	63.5	53.7	40.2	27.2
		MCDB	32.0	41.9	55.9	68.5	75.7	79.3	82.9	80.2	74.3	63.0	49.5	34.6
	10%	WB	22.5	29.5	39.1	50.0	58.0	65.5	71.1	69.6	61.8	51.1	37.5	25.1
		MCDB	28.9	38.5	52.4	64.7	73.5	77.9	81.0	78.3	72.2	59.9	46.4	31.7
Mean Daily Temperature Range	5% DB	MDBR	16.5	18.3	20.1	21.4	21.6	19.9	17.3	17.4	19.2	18.5	16.4	15.6
		MCDBR	19.7	22.2	26.4	28.3	27.9	25.4	22.3	21.1	23.7	23.4	20.7	18.9
		MCWBR	14.0	14.2	14.5	13.2	11.1	8.6	7.7	8.1	9.8	11.9	12.5	13.0
	5% WB	MCDBR	19.0	20.9	24.8	23.7	21.9	19.5	16.8	16.1	18.5	19.2	19.1	18.3
		MCWBR	13.7	13.6	14.1	11.9	9.8	7.6	7.3	7.3	9.0	10.9	11.9	12.8
Clear-Sky Solar Irradiance	taub		0.317	0.390	0.419	0.483	0.537	0.628	0.598	0.583	0.508	0.445	0.354	0.325
	taud		2.227	2.029	2.000	1.910	1.808	1.679	1.789	1.799	1.908	1.992	2.194	2.233
	Ebn at Noon		260	252	260	252	241	219	224	222	230	229	242	246
	Edh at Noon		30	44	50	59	67	76	68	65	55	45	31	28
All-Sky Solar Radiation	RadAvg		824	1109	1520	1806	2017	1855	1780	1694	1448	1151	847	713
	RadStd		55	69	114	115	137	148	97	108	82	75	87	54

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	-2.43	N/A	N/A	N/A	-0.74	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.67	N/A	N/A	N/A	N/A

Nomenclature: See separate page